

MADHURI POKALA

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SENIOR SOFTWARE ENGINEER - FULL STACK

Senior Software Engineer with 9+ years of experience building scalable, high-performance web applications and distributed systems. Strong expertise in React, TypeScript, Node.js, and NestJS, with a deep focus on frontend architecture, API design, and performance optimization. Proven track record of owning features end-to-end—from system design through production deployment—while guiding engineers, driving technical decisions, and delivering reliable, user-centric product experiences at scale.

SKILLS

Programming Languages & Databases: JavaScript, TypeScript, Python, C#, MongoDB, PostgreSQL

Frameworks & Libraries: React, Node.js, Express.js, NestJS, Tailwind CSS, REST API, GraphQL, Prisma, JWT, OAuth, Okta, Pub/Sub, Cypress, Vitest

Cloud & DevOps: AWS (EC2, S3, ECS, RDS, ALB, NLB, ECR), Docker, Kubernetes, Terraform, Jenkins. CI/CD

Tools & Practices: Git, Turborepo, pnpm, ESLint, Biome, Vite, Postman, Veracode, SonarQube, AppSpider, TDD, Agile, Design Patterns, Code Reviews, Confluence, JIRA.

AI & Emerging Technologies: GenAI, LLM, RAG, LangChain, LangGraph

WORK HISTORY

Experian

Senior Software Engineer

03/2023 - Present (California, USA)

01/2021 - 02/2023 (Hyderabad, India)

Global Architecture & Scale

- Architected a monorepo ecosystem (Turborepo / pnpm) with automated semantic versioning (Changesets) and file streaming infrastructure (S3 Presigned URLs); achieved 75% code reuse across US/EU/APAC and supported 3GB+ uploads with 40% lower latency.
- Spearheaded monolith-to-microservices migration with automated PostgreSQL schema management, ensuring data consistency and minimal disruption across global regions.
- Designed region-aware i18n architecture (i18next + PostgreSQL), enabling dynamic content retrieval for localized experiences across all regions.

Performance & Innovation

- Modernized frontend infrastructure with Vite, resolving CommonJS/ESM issues to reduce build times by 60%; redesigned Ops UI driving 75% increase in user engagement.
- Prototyped a region-aware RAG chatbot (LangChain+LLM) embedded in Ascend Ops, providing contextual, self-service guidance for app navigation and error recovery across US/EU/APAC; reduced support tickets by 30% in POC

Security, Reliability & Leadership

- Engineered backend auth (NestJS Guards, OAuth 2.0) with granular RBAC; integrated SonarQube/Veracode into CI/CD, reducing vulnerabilities by 80%.
- Established observability (Prometheus) and testing (Cypress/Vitest), achieving 92% coverage and reducing MTTR by 70%.
- Mentored 4+ engineers and collaborated with product leadership to define technical roadmaps, delivering high-impact features that aligned with quarterly business goals.

GEP - Hyderabad
Software Engineer

04/2020 - 12/2020

- Built full-stack modules for Self-Service Portal and Bill of Materials using React and Node.js, delivering scalable UI workflows and REST APIs aligned with evolving product requirements.
- Collaborated with product stakeholders during sprint planning to translate business requirements into technical designs and acceptance criteria, ensuring predictable feature delivery.
- Designed reusable frontend components and optimized client-side rendering to improve maintainability and performance of enterprise procurement workflows.

Accenture - Hyderabad
Software Engineer

05/2017 - 02/2020

- Developed enterprise-grade web applications using React, ASP.NET Core, and MongoDB, integrating secure authentication flows (OAuth2, JWT, Okta) and Azure services including Key Vault and Web Jobs.
- Strengthened application security posture by resolving Veracode findings and implementing secure coding practices across API and frontend layers.
- Implemented performance monitoring and telemetry using Azure Application Insights, proactively identifying and resolving production issues to improve system reliability.

EDUCATION

Sreenidhi Institute of Science and Technology, Telangana, India
Bachelor of Technology - EEE

05/2013 - 05/2017
